Kazi Fahim Tahmid

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Research Interest

Enhancing Model Compression methods and Reasoning skills to facilitate knowledge distillation while creating specialized SLMs, particularly aimed at biomedical devices, that can be deployed on end-devices, emphasizing on low-resource constraints for edge computing, ensuring privacy in medical applications, and integrating multimodal learning to bring forth precise and dependable real-time surgical and diagnostic automation.

Core Topics

- Model Compression Techniques for SLM Development
- Multimodal Reasoning in SLMs for Biomedical Data
- Machine Learning for Clinical Health and Computational Biology
- Deep Learning for Biomedical Imaging and Disease Diagnostics
- Development and Finetuning of Foundation Models within SLM scope for Multimodal Biomedical Applications

Education

Khulna University of Engineering & Technology Khulna, Bangladesh
B.S. in Computer Science and Engineering Jan 2020 – Sept 2025

CGPA: 3.86/4.00 (Graduated as top 6% in class)

- **Advisor:** Professor K. M. Azharul Hasan
- **Senior-year Thesis Supervisor:** Professor Sk. Imran Hossain

Awards & Grants

Dean's List Award 2021,2022,2023,2024
KUET Faculty of Electrical and Electronics Engineering

- Receipent of Fresher, Sophomore, Junior years for respectively Sessions 2019-2022
- Nominated for Senior year for Session 2022-23

Academic Excellence Technical Grant 2022,2023,2024
KUET Department of Computer Science and Engineering

- Receipent for Sophomore, Junior and Senior year for Sessions 2020-23

Research Experience

Intelligent Computing Lab — CSE, MIST
Research Associate — October 2025 – Present

Responsibilities and Experience

- Developed Activation-Aware Gradient-Based Quantization with Distillation where used joint optimization of activation quantization and distillation via gradient flow analysis to reduce activation error by a significant margin

K-MINDS — CSE, KUET
KUET Machine INtelligence and Deep learning Society — May 2025 – Present
Research Associate

Responsibilities and Experience

- Designed RePlan-SLM, An RL-Enhanced Small Language Model for Efficient Reasoning, Tool Orchestration, and DAG-Based Task Execution
- Formulated Vision-RAG in the hyperbolic domain specifically using Poincaré disk models, to capture hierarchy better in the different radius distance from the center of the sphere [collaboration with Intelsense.AI]

Computational Intelligence & Research Group — CSE, KUET
Undergraduate Research Assistant — Feb 2024 – Sept 2025

Senior-year Research Thesis
Supervised by: Professor Sk. Imran Hossain

Responsibilities and Experience

- Formulated B-GFAN: Bayesian-Graph Fourier Analysis Network, a novel uncertainty-aware Lightweight Model with proper hierarchy modelling and learnable spectral frequencies through fourier co-efficients and angular frequencies for Epilepsy Detection achieving 7x model compression (4.2M) along with better accuracy (97.8%) and F1 score(94.7%) accross patients with well-calibrated uncertainty quantification (ECE=3.1%)
- Designed Attention-Augmented ConvNeXt-BiLSTM Dual-Branch Architecture for Hybrid EEG-Based Epileptic Seizure Detection achieving real-time (4.64 ms inference time) and achieving accuracy up to 95.76% with an F1-score of 91.25% in LOSO validation with 28M parameter count

Manuscripts in Preparation

1. Senior-year Research Thesis: Bayesian-Graph Fourier Analysis Network: Lightweight Model for Epilepsy Detection and Comparison with Attention-Enhanced ConvNeXt-BiLSTM
extending for IEEE Transaction submission

2. NeurIPS 2026 Submission: Position: From Visual Plausibility to Use-Bound Validity - Governing Synthetic Data in Medical AI
Working on rebuttal for ICML 2027 submission

3. From Weight-Space Programming to Adversarial Weight Surgery and Learned Meta-Optimizers
Preprint | November, 2026

Teaching Experience

United International University — Dhaka, Bangladesh
Department of Computer Science and Engineering — July 2026 – Present
Lecturer

Core Courses Conducted

- Artificial Intelligence (CSE 3811)
- Artificial Intelligence Laboratory (CSE 3812)
- Machine Learning (CSE 4889)
- Data Mining (CSE 4891)

Military Institute of Science & Technology — Dhaka, Bangladesh
Department of Computer Science and Engineering — November 2025 – June 2026
Lecturer

Core Courses Conducted

- Structured Programming Language Sessional (CSE 106): Conducted lab sessions on fundamental topics such as Conditional Logic, Iterations, Function, User-Defined Data Structures for freshers to build a introductory project of a real-life problem solver and graded the final Project Submission
- Data Structures and Algorithms - II (CSE 215 216): Implemented advanced Graph algorithms, Dynamic Programming, and Greedy techniques in lab sessions and designed assignments on optimization of network flows and complex data structure efficiency
- Digital System Design Sessional (CSE 316): Instructed students on Register-Transfer Logic, Control Logic Design for them to learn computer design and designed assignments on Barrel Shifter, SAP-1 and SAP-2 (teamwise unique instruction set)
- Software Development Project-I (CSE 364): Demonstrated Software Design Life Cycle (SDLC), UI/UX Design and Full-stack Development procedures in lab sessions for students to present a functional and deployable application as final project
- Pattern Recognition Sessional (CSE 444): Conducted lab session on estimation techniques e.g. Classification, Regression, Clustering, Decision tree to identify different types of pattern in data

Workshop

- Deep Dive into Deep Learning (D3L): Introduced and conducted a 2 week-long intense theoretical and corresponding hands-on coding workshop on **RNN, Transformers, Attention Mechanism, CNN, Deep Generative Modeling, RL and LLMs** for kick-starting Junior-Year students' research journey

Advising

- Actively advising 11 Undergraduate students of Junior Year about **personal, academic and financial** aspects to help them thrive
- Initiated and arranged 5 focus group sessions for students of any discipline to enable **speaking out about the mental health issues they are facing** and have group discussions to find a solve

K-MINDS CSE, KUET

Founding Senior Instructor May 2025 – Sept 2025

- Mentored Undergrad students of KUET on **CoreML** and **Reinforcement Learning** training for Senior-year Thesis advancements
- Conducted workshops on development of **compressed** and **multimodal LLMs** from scratch for end-device deployment
- Organized and managed lab session on **Model Compression** and working with **Privacy-preservation techniques** for mentees to get hands-on experience with relevant frontiers such as LoRA, Quantization, Knowledge Distillation

Industry Experience

Intelsense.AI Dhaka, Bangladesh

Software Engineer July 2025 – Present

Responsibilities and Experience

- End-to-End ASR, TTS Development, Fine-tuning, Scaling
- Handling fast inference and load balancing for local hosting
- Agentic AI, RAG Pipelines, Agent Tool Dev & Deployment

Products Developed

- UnisenseAI, a one-stop-solution for an unified response system paired up with complete lead and transaction automation with large-scaled recommendation models
- SenseForm, a central privacy-protected form automation system for Bangladesh's national banking for home-banking and navigation with multilingual support
- AirVoice, a ASR-TTS coupled system developed for improving Huawei's Customer Service Op

Leadership

BitFest 2025 Khulna, Bangladesh

Organizer Jan, 2025

- Organized Datathon, Hackathon, LFR, Soccerbot, Poster Presentation competition segments for 12000+ undergraduate students from 150+ universities
- Participated in the dataset and task development team for preparation of Datathon and Hackathon segments
- Represented the senior undergraduate Judge in the jury panel for Hackathon and Poster Presentation

KUET Inter University Programming Contest 2025 Khulna, Bangladesh

Co-ordinator Jan, 2025

- Successfully organized an inter-university programming contest, managing participation from over 150 teams of different universities across the country.

9th ISTARC Science & Technology Festival Khulna, Bangladesh

Festival Organizer Sept, 2016

- Coordinated a national high school science festival, engaging over 5,000 students in various activities, including Olympiads, project exhibitions, and gaming competitions

Academic Projects

CastleBlack
OpenGL, C++
- Created a 3D medieval castle in OpenGL
- Used Bezier curves, fractal trees, and dynamic lighting
- Custom shader programming for real-time rendering and procedural generation

LangBridge
Flex, Bison, C++
- Designed a custom programming language using Flex and Bison
- Included lexical, syntax, and semantic analysis

InpainterPro
Python, OpenCV
- Implemented an optimized Telea method for structure and texture reconstruction
- Targeted restoration of missing or damaged image regions

CoinScribe
Swift
- Developed a crypto wallet tracker app in Swift
- Included features for note-taking and to-do list management
- Included real-time currency exchange track charts with historical data forecasting of the stock market to make it a one-stop decision-making platform

Technical Skills

Compression Quantization (INT8/FP16), Pruning (Structured/Unstructured), Knowledge Distillation, Low-Rank Factorization, Model Profiling & Benchmarking

Python PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, NumPy, Pandas

MultimodalAI Vision-Language Models, Cross-Modal Fusion, Multi-Modal Transformers, Attention Mechanisms

MLOps Docker, K8s, Model Versioning, CI/CD Pipelines, Performance Monitoring, A/B Testing

Alignment SFT (Supervised Fine-Tuning), DPO (Direct Preference Optimization), Knowledge Distillation for Alignment, Safety-Preserving Compression

HealthcareAI Medical Data Preprocessing, HIPAA-Compliant Systems, Clinical Dataset Handling, Privacy-Preserving ML

English Proficiency

IELTS Overall Band: 8.0

Academic